

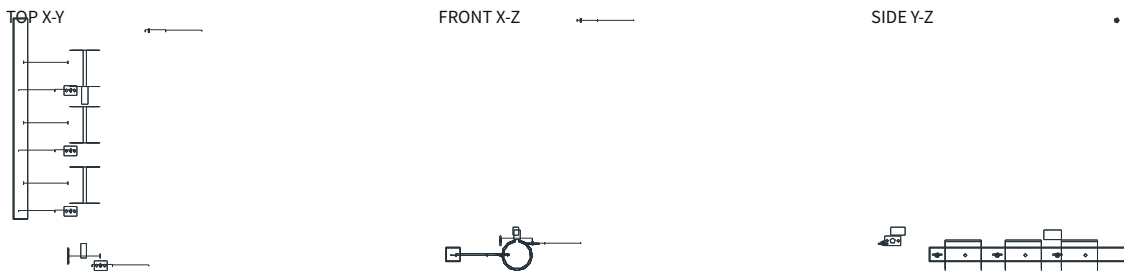
1 EX-003 — heater/thermocouple layout

물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact received component 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

전체 배치: GGM-FULL-ASM / 구동계: GGM R2. 기본 조립 투상도는 통합 전 참조이며 최신 전체 배치가 아니다. Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

EX-003 heater thermocouple

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 19



Overall X 234.40 · Y 313.50 · Z 315.07 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

1.1 제작·검사 기준

- 단위: mm · 제3각법 · 축척: NTS(기입 치수 우선)
- 재료/구성: mica/NiCr heater and thermocouple candidates; TH-INS-01 secondary shield-exterior barrier after S4 qualification
- 핵심 치수/공차: Band free-state ID34.10–34.20 and usable closure ≥ 1.00 leave worst-case closure reserve0.277 mm; after cold tightening, 0.05 mm feeler penetration ≤ 5 mm at 8 sectors excluding split $\pm 10^\circ$ is physical NOT_RUN. HOLD T1–T3: probe unselected; OD tolerance and insertion stop unqualified. Barrel flat-bottom bore 5.40 ± 0.05 preserves conservative ligament $3.345 \geq 3.32$, but does not qualify probe insertion; do not deepen or force probe to bottom. T4 insertion10/blind12 and T5 insertion4 are nominal candidates. Qualify retention, insulation and thermal response before assembly. TH-INS-01 is allowed only on the outside face of the grounded metal hot shield and must not cover heaters, barrel/die, cutoffs, probes, terminals or vents; S4 coupon smoke and P9 interface-temperature margin remain physical gates. SYS-04 die joint digital load-path PASS at1.50 N·m; physical receipt/leak/first thermal cycle NOT_RUN; shield clearance requires service verification.

- 일반공차: ISO 2768-m. 개별 부품은 exports/final/manufacturing 도면과 interface_catalog.csv를 함께 검사한다.
- Source: b287f07f7a55ba61e39c0a5cbe8e45515c9e53ec · 물리 검증: NOT_RUN